

MOSTAFA HANY

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EXPERIENCE

Software Engineer

Unify

Feb 2025 - Apr 2026

| Remote

- Owned the development and maintenance of the Python SDK for the Unify API.
- Increased logging throughput by $10\times$ by redesigning the pipeline using asynchronous (asyncio-based) processing, reducing overhead in high-throughput environments.
- Reduced system initialization time by $4\times$ through improved dependency loading strategies.
- Built and extended caching systems, including semantic caching for LLM interactions, reducing redundant requests, latency, and compute cost.
- Implemented context compression and time-awareness mechanisms for long-running LLM workflows, enabling efficient handling of extended conversations within token constraints.
- Refactored core agent execution loop into a modular, maintainable architecture, enabling easier extensibility and faster iteration on new features.
- Diagnosed and resolved nondeterministic prompt behavior that led to large-scale cache inconsistencies, improving correctness and cache efficiency.
- Expanded multi-provider LLM support by migrating to a LiteLLM based abstraction layer.
- Developed a cost tracking infrastructure for LLM usage, improving visibility and control over API spend.
- Established a testing infrastructure (custom fixtures and decorators) to improve the reliability, debuggability, and execution speed of test suite.
- Designed and implemented end-to-end tracing and metrics for LLM requests, improving observability, debugging speed, and system reliability.

Software Engineer (Computational Geometry)

Atomica

Sep 2024 - Feb 2025

| Remote

- Created comprehensive visual examples to demonstrate library usage, enabling developers to easily understand and effectively utilize the API.
- Enhanced CI workflows to ensure more stable releases and a streamlined development process.
- Integrated run-time sanitizers to enhance code reliability and debug various issues.
- Incorporated static analysis tools into the code base, significantly improving code quality and reliability.
- Optimized core functions, improving execution time from 5s to 1s, leading to a $5\times$ speedup.

Software Engineer

Unify

Aug 2022 - Jan 2024

| Remote

- Managed a team of five engineers.
- Designed and implemented a proof-of-concept for a source-to-source transpiler to convert code between different deep learning frameworks, e.g., TensorFlow to PyTorch.
- Redesigning the testing framework to improve modularity and readability, achieving a 50% reduction in code size and making code multiprocessing friendly.
- Designed and implemented functionality to support the import of multiple instances of the framework with different backends at runtime, reducing the average test runtime from 120 to 10 seconds.
- Designed and implemented an API backend generation feature using Python AST (Abstract Syntax Trees), allowing developers to generate a new API backend with only a few lines of configuration.
- Extended the integration of property-based testing in the testing framework to specifically meet our requirements with less runtime overhead and reduced code.
- Extended and maintained the Ivy framework support on Windows.
- Added support for GPU testing in the testing framework.

PUBLICATIONS

Accelerating Deep Learning using Ivy | Link

NeurIPS, 2023

Guillermo Sanchez-Brizuela, Ved Patwardhan, Matthew Barrett, Paul Anderson, **Mustafa Hani**, Daniel Lenton.

RELATED SKILLS

Programming Languages: C++, C, GLSL, Python, Lua, C#

Libraries / Frameworks: Vulkan, OpenGL, raylib, WebGL, Unity, ImGui, SFML, Assimp, GLM, GLFW, three.js

Dev Tools: RenderDoc, CMake, Tracy, Premake, Docker, git, gdb

EDUCATION

Faculty of Computers & Artificial Intelligence, Cairo University

Giza, Egypt

Bachelor of Science in Computer Science, Artificial Intelligence

Sep 2024

PERSONAL PROJECTS

Ptah (3D Renderer) | Link

C++, OpenGL, ImGui, Assimp, GLAD, CMake

a real-time 3D rendering engine in featuring a forward pipeline with Lambert and PBR shading, a hot-reloading shader preprocessor, and a post-processing stack (bloom, HDR, gamma correction, anti-aliasing).

Software Raytracer | Link

C++, Lua C API, STB, Premake

Cross-platform software raytracer from scratch in C++. Supports lighting, shadows, and reflections. Embedded Lua C API for scene configuration using a Lua file.

Shadowmap Visualizer | Link

JS, WebGL2, gl-matrix

An interactive WebGL2 demo for understanding how shadow mapping works.

The Office (Game) | Link

C#, Unity

Brackey's Game Jam 2021 submission. Built with Unity in 6 days.

Gumballs Rush (Game) | Link

C#, Unity

Built and shipped a hyper-casual mobile game in Unity, covering gameplay mechanics, physics, and UI. published live on the App Store.

BingoBango (Game) | Link

C#, Unity

A solo-developed 3D vertical platformer released on Steam, featuring a custom character controller with physics-based jump and momentum, five difficulty levels, and Steamworks integration.

CS:GO Object Detection | Link

Python, YoloV5, PyTorch, OpenCV

Detecting CS:GO objects using YOLOv5 with a custom OpenCV annotation script.